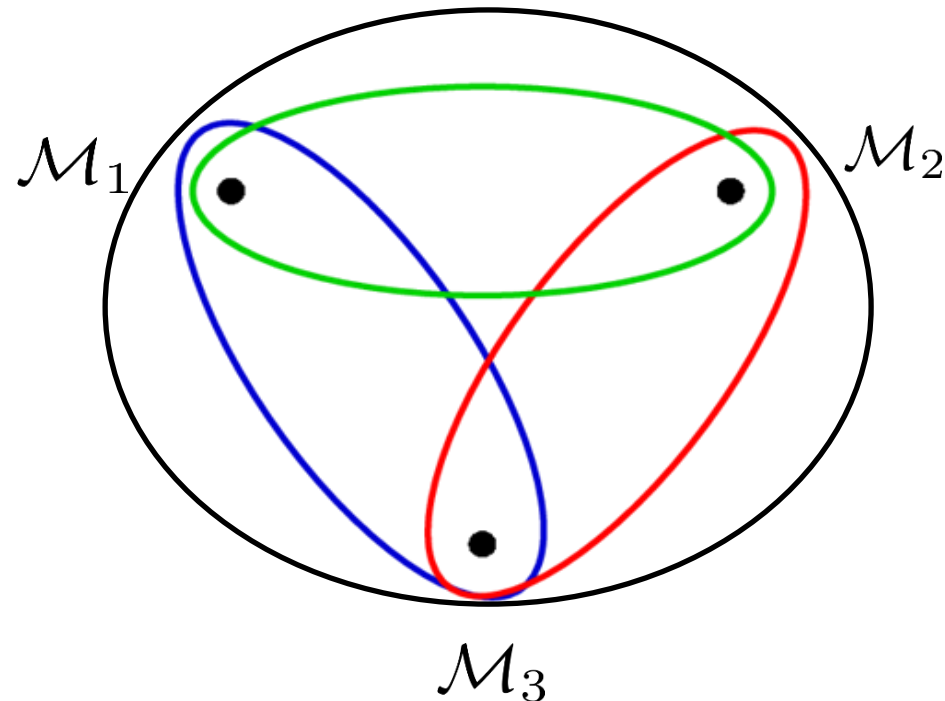


Distributed Quantum Incompatibility

Lucas Tendick, Hermann Kampermann, and Dagmar Bruß

PRL **131**, 120202 (2023)

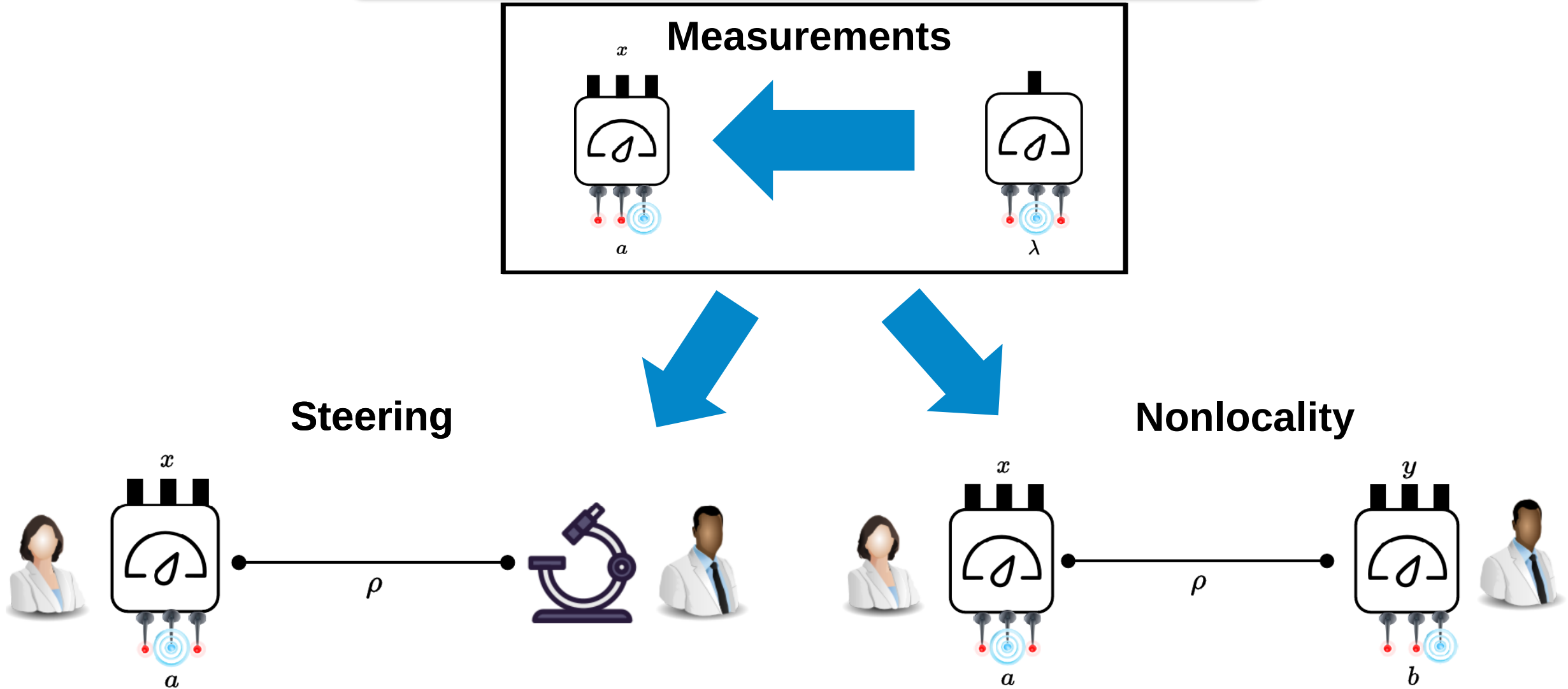
arXiv: 2301.08670



Inria

hhu Heinrich Heine
Universität
Düsseldorf

Quantum Incompatibility



Distribution of Quantum Resources

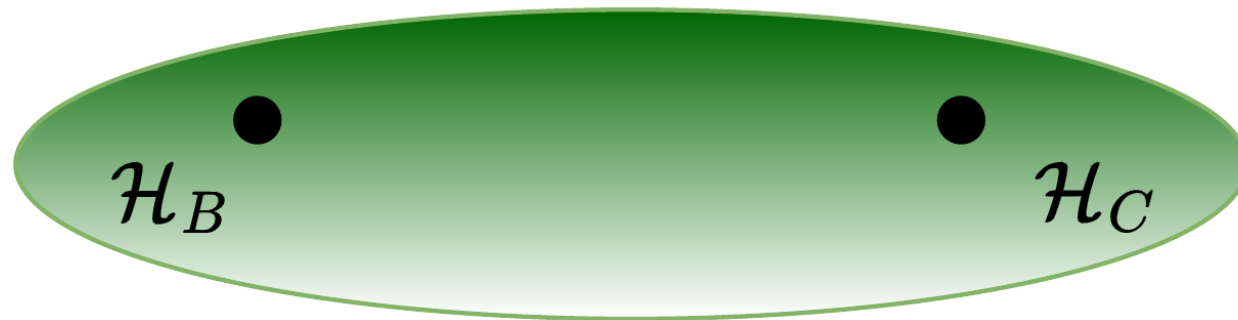
„Distributed
entanglement“

PRA **61**, 052306
(2000)

● $\mathcal{H}_A$

C_{AB}^2

$$C \leq E_{A|BC}$$



Coherence

„Monogamy of Bell
correlations and
Tsirelson's bound“

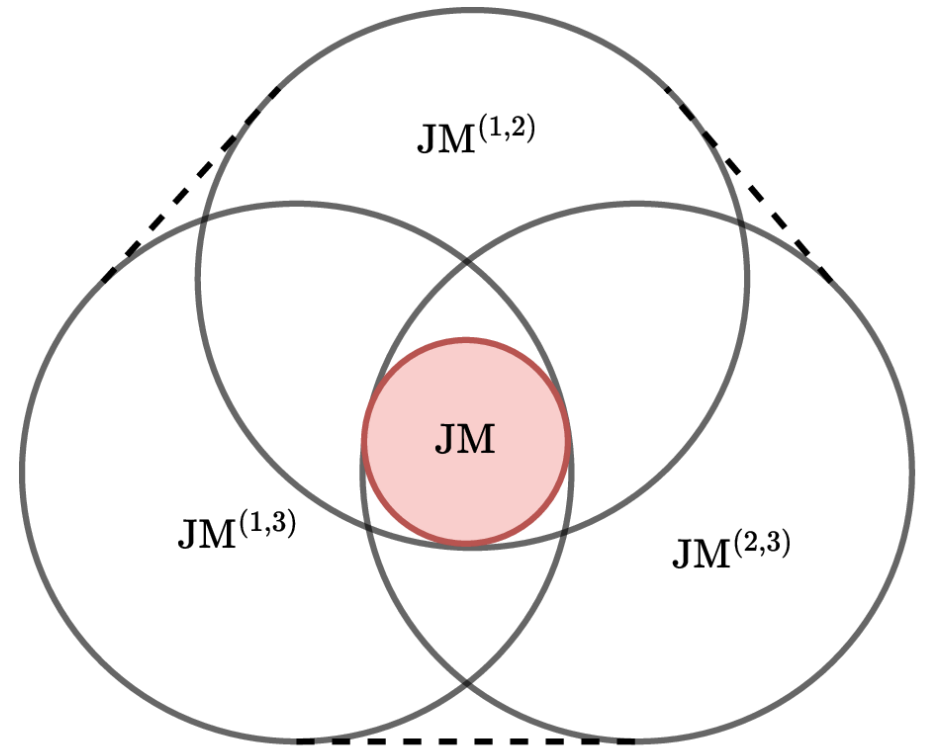
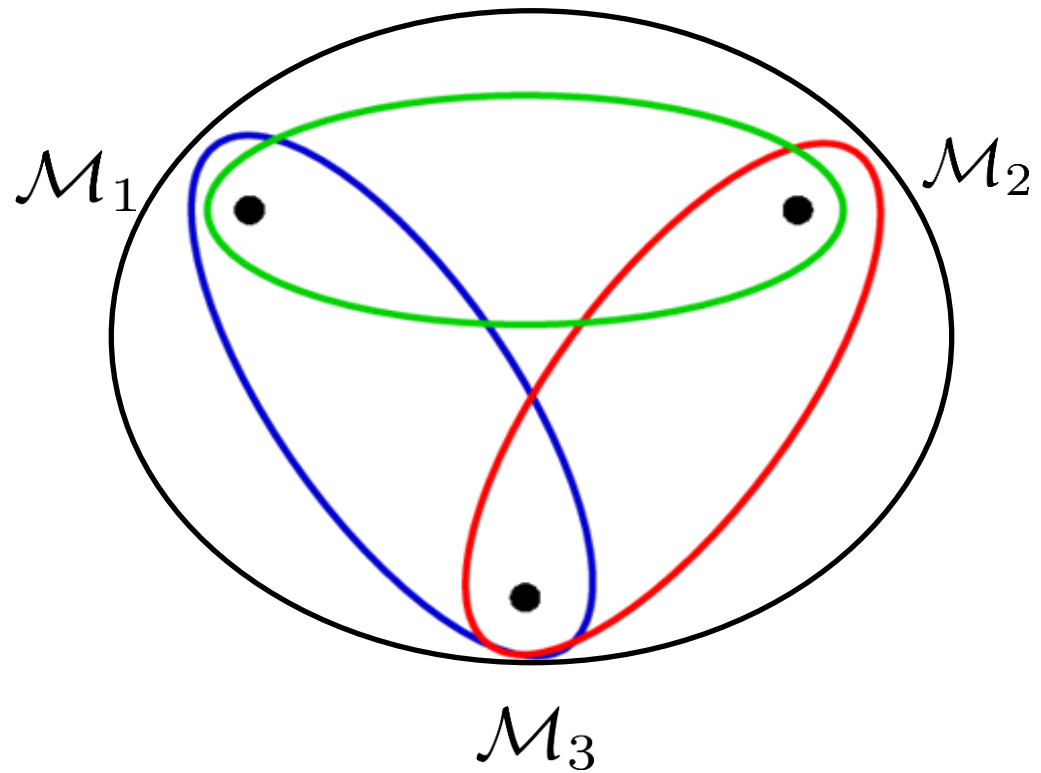
arXiv:quant-
ph/0611001

„Distribution of Quantum
Coherence in Multipartite
Systems“

PRL **116**, 150504 (2016)

$$\mathcal{C} \leq \mathcal{C}_{\text{local}} + \mathcal{C}_{\text{global}}$$

Distribution of Measurement Incompatibility



Measurement Incompatibility

QM Lecture

Heisenberg-Robertson uncertainty relation

$$\Delta_\psi A \cdot \Delta_\psi B \geq \frac{1}{2} |\langle [A, B] \rangle_\psi|$$

Noncommutativity of observables

$$[A, B] = AB - BA \neq 0$$

Observables $\leftrightarrow$ Projective measurements

$$A = \sum_a \mu_a P_a \longrightarrow \{P_a\}$$

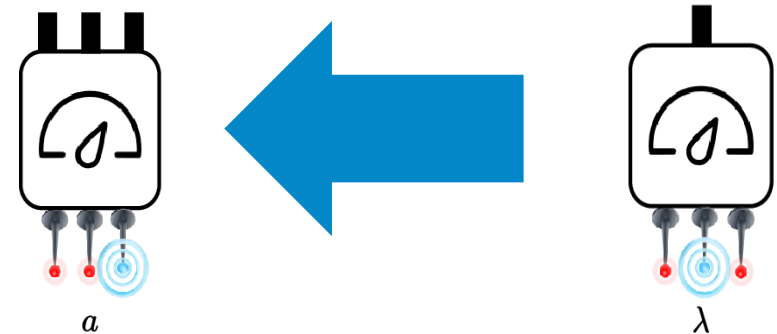
$$B = \sum_b \nu_b Q_b \longrightarrow \{Q_b\}$$

- Limitation
- 2 Measurements

Quantum information

Joint measurability

$$M_{a|x} = \sum_\lambda p(a|x, \lambda) G_\lambda \iff M_{a|x} = \sum_{\vec{a} \setminus a_x} G_{\vec{a}}$$



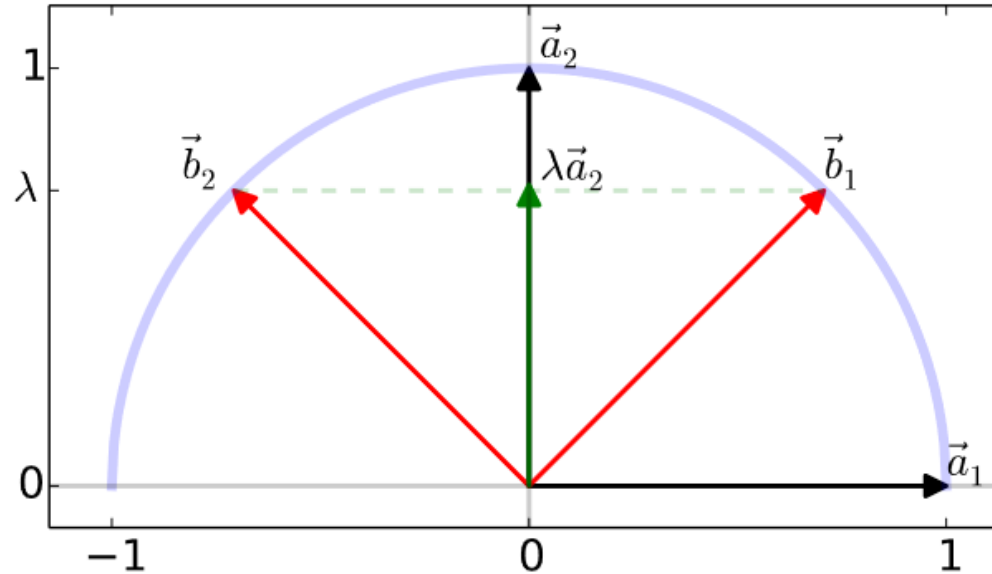
Generalization of noncommutativity to POVMs

- Resource for quantum information processing tasks
 - State discrimination/exclusion
 - Cryptography
 - Bell nonlocality and steering

T. Heinosaari et al. J. Phys. A: Math. Theor. **49**, 123001 (2016)

O. Gühne et al. Rev. Mod. Phys. **95**, 011003 (2023)

Incompatibility of two Measurements



Uola et al. Phys. Rev. A 94, 022109 (2016)

Marginals

$$M_{\pm|1}^{\lambda} = \sum_t G_{s,t}^{\lambda}$$

$$M_{\pm|2}^{\lambda} = \sum_s G_{s,t}^{\lambda}$$

POVM

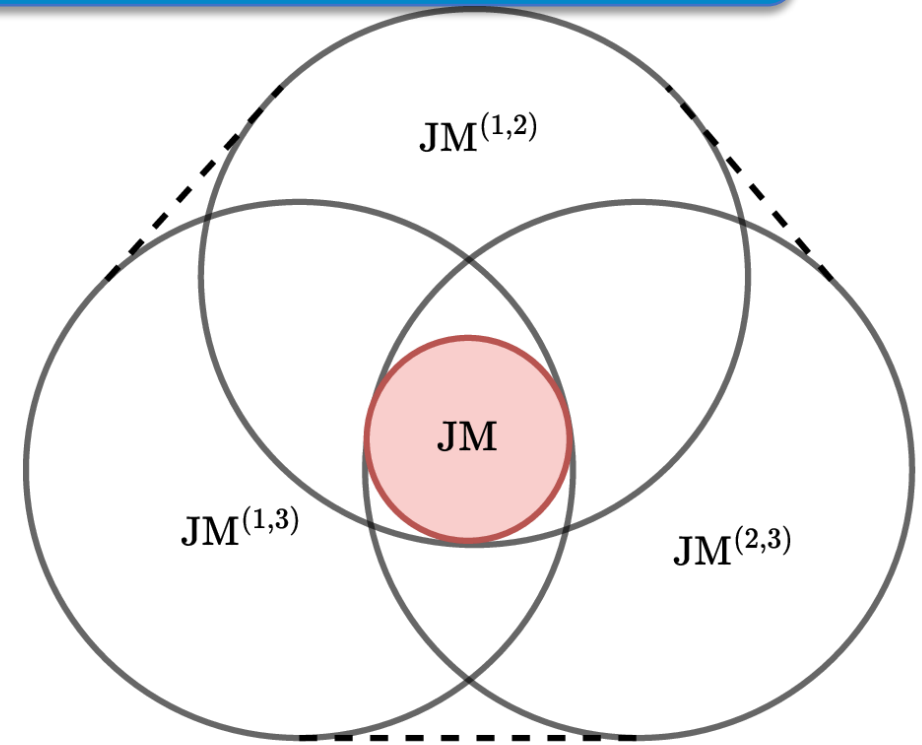
Completeness: ✓

Positivity: $\lambda \in [0, 1/\sqrt{2}]$

- Jointly measurable (JM) for $\lambda \in [0, 1/\sqrt{2}]$
- Jointly measurable but noncommuting

Incompatibility of multiple ($\equiv$ three) Measurements

- **Assemblage:** $\mathcal{M}_{(1,2,3)} = (\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3)$
- **JM:** $M_{a|x} = \sum_{\lambda} p(a|x, \lambda) G_{\lambda}$
- **JM pairs:** $\text{JM}^{(1,2)}, \text{JM}^{(1,3)}, \text{JM}^{(2,3)}$
- **Pairwise:** $\text{JM}^{\text{pair}} := \text{JM}^{(1,2)} \cap \text{JM}^{(1,3)} \cap \text{JM}^{(2,3)}$
- **Genuine triplewise:** $\mathcal{M}_{(1,2,3)} \notin \text{Conv}(\text{JM}^{(1,2)}, \text{JM}^{(1,3)}, \text{JM}^{(2,3)})$
- **Noisy Pauli X, Y, Z measurements:** $M_{\pm|i}^{\eta} = \frac{1}{2}(\mathbb{1} \pm \eta \sigma_i)$
 - **JM:** $\eta \leq 1/\sqrt{3} \approx 0.58$
 - **Pairwise:** $\eta \leq 1/\sqrt{2} \approx 0.71$
 - **Genuine triplewise incompatible:** $\eta \geq \frac{\sqrt{2}+1}{3} \approx 0.8$



Different incompatibility structures result in different restrictions on Bell inequality violations

M.T. Quintino et al. PRL 123, 180401 (2019)

Surprisingly little is known...

Numerically

Bavaresco et al. PRA **96**, 022110 (2017)

Projective qubit measurements						
Gen. opt.		Planar opt.		Fixed sets		
N	Upper	Lower	Upper	Lower	Thomson	Fibonacci
2	0.7071	0.7071	0.7071	0.7071	0.7071	0.7102
3	0.5774	0.5755	0.6667	0.6667	0.5774	0.6981
4	0.5547	0.5437	0.6533	0.6532	0.5774	0.6114
5	0.5422	0.5283	0.6472	0.6470	0.5513	0.5653
6	0.5270		0.6440		0.5393	0.5561
7	0.5234		0.6420		0.5234	0.5533
8	0.5202		0.6407		0.5250	0.5508
9	0.5149		0.6399		0.5209	0.5359
10	0.5144		0.6392		0.5191	0.5302
11	0.5132		0.6388		0.5148	0.5274
12	0.5117		0.6384		0.5152	0.5261
13	0.5105		0.6382		0.5126	0.5220
14			0.6380		0.5114	0.5180
15			0.6378		0.5107	0.5158
16					0.5106	0.5158
17					0.5086	0.5150
18					0.5079	0.5136

Analytically

Designolle et al. PRL **122**, 050402 (2019)

MUBs

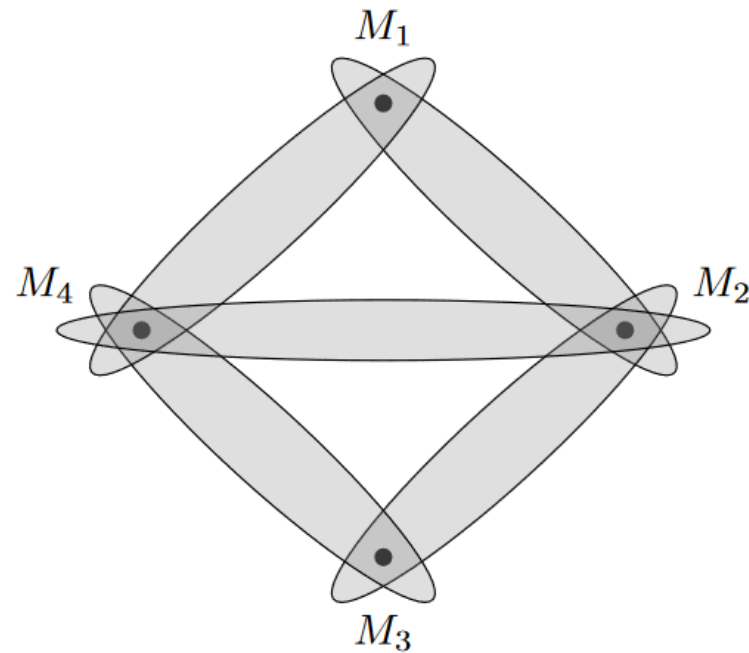
$k \backslash d$	2	3	4	5	6	7
	$\eta^* = \eta_{\text{up}}$	$\eta^* = \eta_{\text{up}}$	$\eta^* \quad \eta_{\text{up}}$	$\eta^* \quad \eta_{\text{up}}$	$\eta^* \quad \eta_{\text{up}}$	$\eta^* \quad \eta_{\text{up}}$
2	$\frac{1}{\sqrt{2}} \approx 0.7071$	$\frac{1+\sqrt{3}}{4} \approx 0.6830$	$\frac{2}{3} \approx 0.6667$	$\frac{3+\sqrt{5}}{8} \approx 0.6545$	$\frac{4+\sqrt{6}}{10} \approx 0.6449$	$\frac{5+\sqrt{7}}{12} \approx 0.6371$
3	$\frac{1}{\sqrt{3}} \approx 0.5774$	$\frac{\cos(\pi/18)}{\sqrt{3}} \approx 0.5686$	0.5469 0.5556	$\frac{1+\sqrt{5}}{6} \approx 0.5393$ $\frac{13-\sqrt{5}+\sqrt{30(5+\sqrt{5})}}{48} \approx 0.5312$	0.5204 0.5254	0.5101 0.5154
4		$\frac{1+3\sqrt{5}}{16} \approx 0.4818$	$\frac{1}{2} = 0.5000$	0.4615 0.4616	? ≤ 0.4550	0.4516 0.4436 0.4488
5			$\frac{3+2\sqrt{3}}{15} \approx 0.4309$	0.4179	?	0.4049 0.4120
6				0.3863	?	0.3754 0.3867
7					?	0.3685
8						0.3318

Surprisingly little is known...

Generally

Kunjwal et al. PRA **89**, 052126 (2014)

Quantum realization of arbitrary joint measurability structures



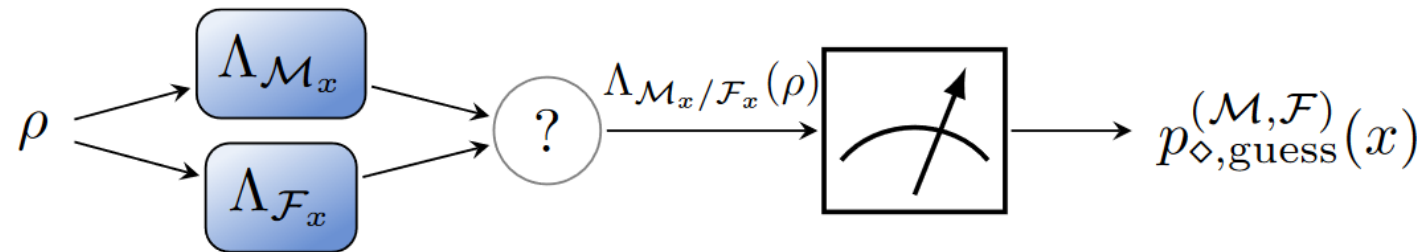
Our Approach: Distance-based Quantification

$$I_{\diamond}(\mathcal{M}_{\mathbf{p}}) = \min_{\mathcal{F} \in \text{JM}} \underbrace{\sum_x p(x) D_{\diamond}(\Lambda_{\mathcal{M}_x}, \Lambda_{\mathcal{F}_x})}_{D_{\diamond}(\mathcal{M}_{\mathbf{p}}, \mathcal{F}_{\mathbf{p}})} \text{ with } \mathcal{M}_{\mathbf{p}} = (\mathcal{M}, \mathbf{p})$$

LT, M. Kliesch, H. Kampermann, and D. Bruß

Quantum 7, 1003 (2023)

- Measure-and-prepare channel: $\Lambda_{\mathcal{M}_x}(\rho) = \sum_a \text{Tr}[M_{a|x}\rho] |a\rangle\langle a|$
- Diamond distance: $D_{\diamond}(\Lambda_1, \Lambda_2) = \max_{\rho \in \mathcal{S}(\mathcal{H} \otimes \mathcal{H})} \frac{1}{2} \|((\Lambda_1 - \Lambda_2) \otimes \mathbb{1}_d)\rho\|_1$ with $\|X\|_1 = \text{Tr}[\sqrt{X^\dagger X}]$

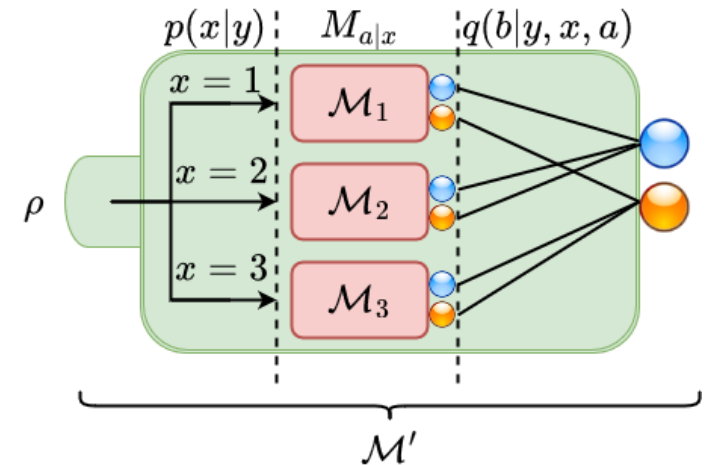


$$\longrightarrow p_{\diamond, \text{guess}}^{(\mathcal{M}, \mathcal{F})}(x) = \frac{1}{2} [1 + D_{\diamond}(\Lambda_{\mathcal{M}_x}, \Lambda_{\mathcal{F}_x})]$$

Distance-based Quantifier: Properties

$$I_{\diamond}(\mathcal{M}_{\mathbf{p}}) = \min_{\mathcal{F} \in \text{JM}} \underbrace{\sum_x p(x) D_{\diamond}(\Lambda_{\mathcal{M}_x}, \Lambda_{\mathcal{F}_x})}_{D_{\diamond}(\mathcal{M}_{\mathbf{p}}, \mathcal{F}_{\mathbf{p}})} \text{ with } \mathcal{M}_{\mathbf{p}} = (\mathcal{M}, \mathbf{p})$$

- Monotone under application of quantum channels $\Lambda^{\dagger}: \mathcal{M} \xrightarrow{\Lambda^{\dagger}} \Lambda^{\dagger}(\mathcal{M})$, s.t. $\text{Tr}[M_{a|x} \Lambda(\rho)] = \text{Tr}[\Lambda^{\dagger}(M_{a|x}) \rho]$
- Monotone under classical simulation $\xi: \mathcal{M} \xrightarrow{\xi} \mathcal{M}'$, $M'_{b|y} = \sum_x p(x|y) \sum_a q(b|y, x, a) M_{a|x}$, $p(x) = \sum_y p(y) p(x|y)$
- Inherits properties of a distance
- Convex combination over single settings
- Efficiently computable via convex optimization $\rightarrow$ Semidefinite programming
- Operational interpretation



A first Incompatibility Bound

$$I_{\diamond}(\mathcal{M}_{\mathbf{p}}) = \min_{\mathcal{F} \in \text{JM}} \underbrace{\sum_x p(x) D_{\diamond}(\Lambda_{\mathcal{M}_x}, \Lambda_{\mathcal{F}_x})}_{D_{\diamond}(\mathcal{M}_{\mathbf{p}}, \mathcal{F}_{\mathbf{p}})} \quad \text{with } \mathcal{M}_{\mathbf{p}} = (\mathcal{M}, \mathbf{p}) \quad p(x) = \frac{1}{m} \quad \forall x$$

1) Triangle inequality

$$I_{\diamond}(\mathcal{M}_{(1,2,3)}) \leq D_{\diamond}(\mathcal{M}_{(1,2,3)}, \mathcal{N}_{(1,2,3)}) + I_{\diamond}(\mathcal{N}_{(1,2,3)})$$

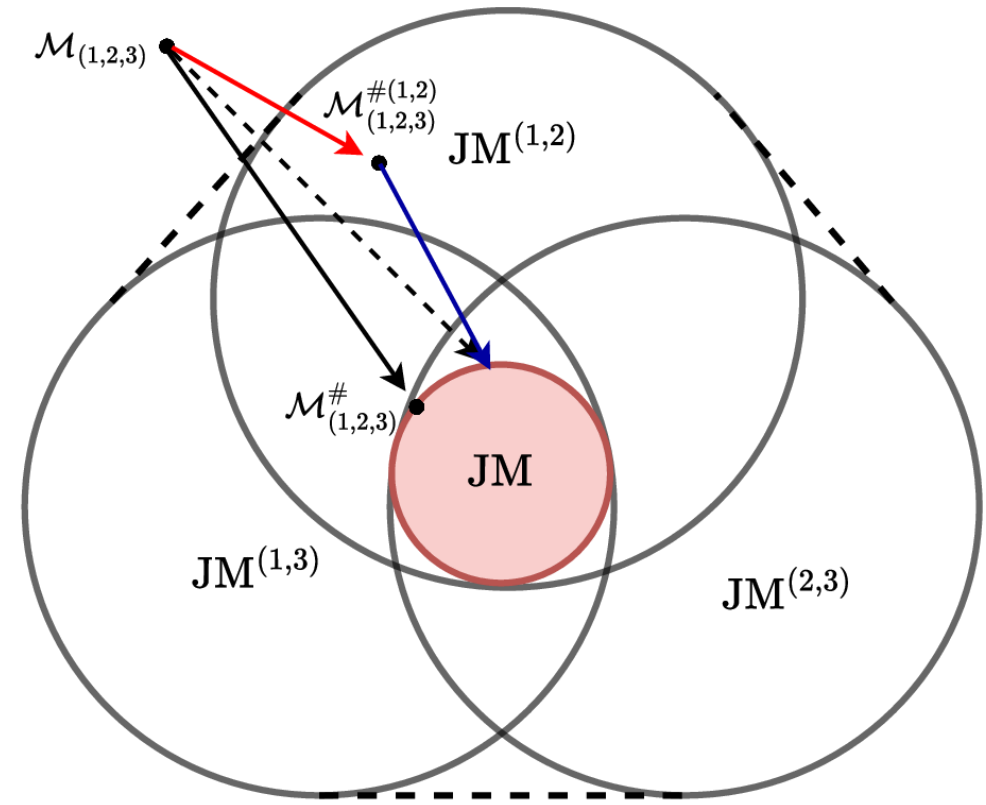
2) Choose clever middle point

$$\mathcal{N}_{(1,2,3)} = \mathcal{M}_{(1,2,3)}^{\#(1,2)} := \mathcal{M}_{(1,2)}^{\#} \mathbin{\sharp} \mathcal{M}_3 = (\mathcal{M}_{(1,2)}^{\#}, \mathcal{M}_3)$$

└ Concatenation of
ordered lists

3) Evaluate

$$I_{\diamond}(\mathcal{M}_{(1,2,3)}) \leq \frac{2}{3} I_{\diamond}(\mathcal{M}_{(1,2)}) + I_{\diamond}(\mathcal{M}_{(1,2,3)}^{\#(1,2)})$$



The Incompatibility gain

1) Measurement splitting

$$\mathcal{M}_{(1,2,3)} = (\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3) \longleftrightarrow \mathcal{M}_{(1,1,2,2,3,3)} = (\mathcal{M}_1, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_3)$$


$$I_{\diamond}(\mathcal{M}_{(1,2,3)}) = I_{\diamond}(\mathcal{M}_{(1,1,2,2,3,3)}) = I_{\diamond}(\mathcal{M}_{(1,2,1,3,2,3)})$$

2) Triangle inequality + clever middle point

$$I_{\diamond}(\mathcal{M}_{(1,2,3)}) \leq \underbrace{\frac{1}{3} [I_{\diamond}(\mathcal{M}_{(1,2)}) + I_{\diamond}(\mathcal{M}_{(1,3)}) + I_{\diamond}(\mathcal{M}_{(2,3)})]}_{\text{Average subset Incompatibility}} + I_{\diamond}(\mathcal{N}) \text{ with } \mathcal{N} = \mathcal{M}_{(1,2)}^{\#} \# \mathcal{M}_{(1,3)}^{\#} \# \mathcal{M}_{(2,3)}^{\#}$$

Note: $I_{\diamond}(\mathcal{N}) \leq I_{\diamond}(\mathcal{G})$ with
 $\mathcal{G} = G(\mathcal{M}_{(1,2)}^{\#}) \# G(\mathcal{M}_{(1,3)}^{\#}) \# G(\mathcal{M}_{(2,3)}^{\#})$

$$\text{Let } I_{\diamond}(\mathcal{M}_{(1,2)}) \geq \max\{I_{\diamond}(\mathcal{M}_{(1,3)}), I_{\diamond}(\mathcal{M}_{(2,3)})\} \implies \Delta I_{(1,2) \rightarrow (1,2,3)} := I_{\diamond}(\mathcal{M}_{(1,2,3)}) - I_{\diamond}(\mathcal{M}_{(1,2)}) \leq I_{\diamond}(\mathcal{N})$$



Again „Polygamous“ behavior of incompatibility

Example I

Noisy Pauli measurements: $\mathcal{M}_{(1,2,3)}^\eta = (\mathcal{M}_1^\eta, \mathcal{M}_2^\eta, \mathcal{M}_3^\eta)$ with $M_{\pm|i}^\eta = \frac{1}{2}(\mathbb{1} \pm \eta\sigma_i)$

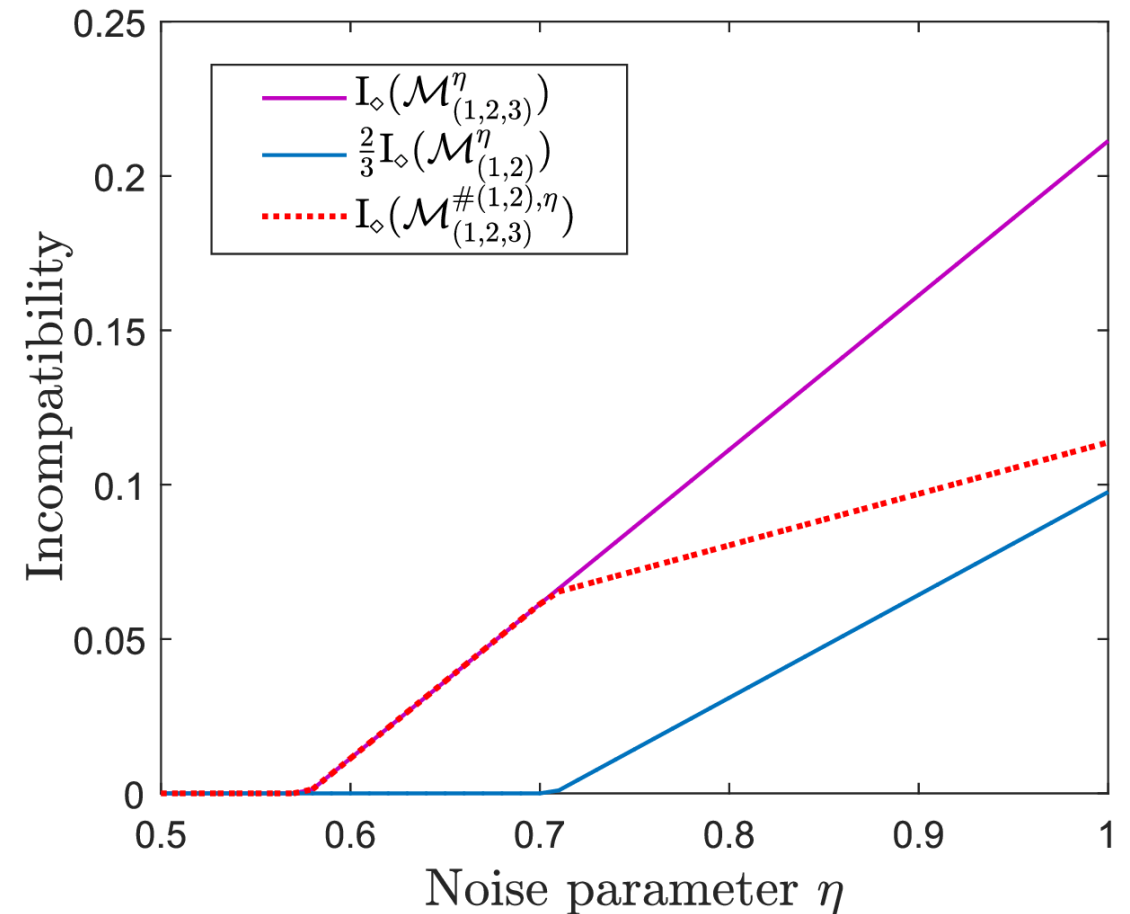
$$\mathcal{M}_{(1,2)}^\eta = (\mathcal{M}_1^\eta, \mathcal{M}_2^\eta) \rightarrow \mathcal{M}_{(1,2,3)}^{\#(1,2),\eta} = \mathcal{M}_{(1,2)}^{\#,\eta} \sharp \mathcal{M}_3^\eta$$

$$I_\diamond(\mathcal{M}_{(1,2,3)}) \leq \frac{2}{3}I_\diamond(\mathcal{M}_{(1,2)}) + I_\diamond(\mathcal{M}_{(1,2,3)}^{\#(1,2)})$$

**Tightness can be shown
analytically!**

Tightness also for the cases:

- $m = 2 \rightarrow m' = d$
- $m = 2 \rightarrow m' = d + 1$
- $m = d \rightarrow m' = d + 1$



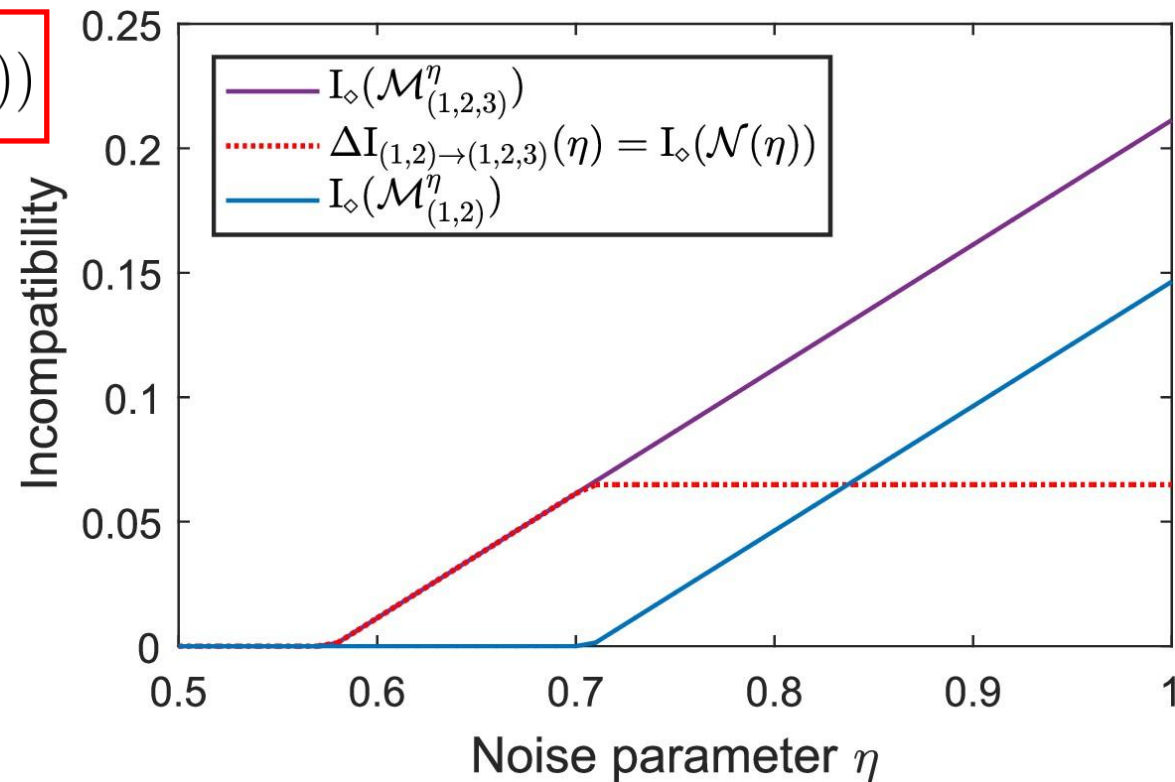
Example II

Noisy Pauli measurements: $\mathcal{M}_{(1,2,3)}^\eta = (\mathcal{M}_1^\eta, \mathcal{M}_2^\eta, \mathcal{M}_3^\eta)$ with $M_{\pm|i}^\eta = \frac{1}{2}(\mathbb{1} \pm \eta\sigma_i)$

$$I_\diamond(\mathcal{M}_{(1,2,3)}^\eta) \leq \frac{1}{3} [I_\diamond(\mathcal{M}_{(1,2)}^\eta) + I_\diamond(\mathcal{M}_{(1,3)}^\eta) + I_\diamond(\mathcal{M}_{(2,3)}^\eta)] + I_\diamond(\mathcal{N}(\eta)), \quad \mathcal{N}(\eta) = \mathcal{M}_{(1,2)}^{\#, \eta} \sharp \mathcal{M}_{(1,3)}^{\#, \eta} \sharp \mathcal{M}_{(2,3)}^{\#, \eta}$$

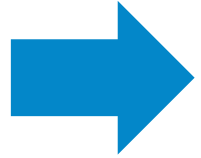
$$\Delta I_{(1,2) \rightarrow (1,2,3)}(\eta) := I_\diamond(\mathcal{M}_{(1,2,3)}^\eta) - I_\diamond(\mathcal{M}_{(1,2)}^\eta) \leq I_\diamond(\mathcal{N}(\eta))$$

**Tightness can be shown
analytically!**



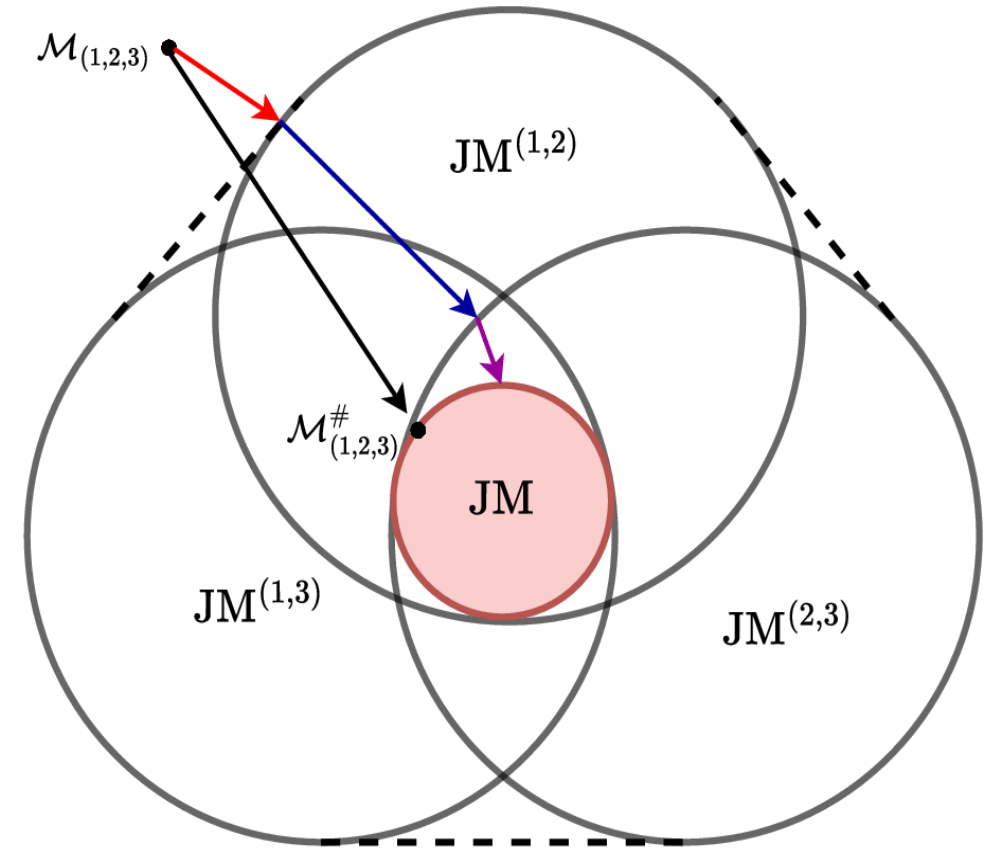
Bounds from Incompatibility Structures

Bound the incompatibility by its incompatibility with respect to different structures



$$I_{\diamond}(\mathcal{M}) \leq I_{\diamond}^{\text{gen}}(\mathcal{M}) + I_{\diamond}^{\text{pair}}(\mathcal{M}) + I_{\diamond}^{\text{hol}}(\mathcal{M})$$

- Follows from the repeated application of the triangle inequality
- Tight for Pauli measurements
- Numerical evidence for tightness of higher dim. MUB measurements



Generalizations

More measurements

$$I_{\diamond}(\mathcal{M}_{(1,2,\dots,m)}) \leq \frac{|C|}{m} I_{\diamond}(\mathcal{M}_C) + I_{\diamond}(\mathcal{M}_{(1,2,\dots,m)}^{\#C})$$

$$I_{\diamond}(\mathcal{M}_{(1,2,\dots,m)}) \leq \sum_i \frac{|C_i|}{m} I_{\diamond}(\mathcal{M}_{C_i}) + I_{\diamond}(\mathcal{M}_{C_1}^{\#} \sharp \mathcal{M}_{C_2}^{\#} \sharp \dots \sharp \mathcal{M}_{C_n}^{\#})$$

$$\Delta I_{(1,\dots,m) \rightarrow (1,\dots,m,m+1)} \leq I_{\diamond}(\mathcal{N}) \leq I_{\diamond}(\mathcal{G})$$

Non-uniform input distributions

$$I_{\diamond}(\mathcal{M}_{(1,2,3)}^{\mathbf{P}}) \leq D_{\diamond}(\mathcal{M}_{(1,2,3)}^{\mathbf{P}}, \mathcal{N}_{(1,2,3)}^{\mathbf{P}}) + I_{\diamond}(\mathcal{N}_{(1,2,3)}^{\mathbf{P}})$$

$$D_{\diamond}(\mathcal{M}_{(1,2,3)}^{\mathbf{P}}, \mathcal{N}_{(1,2,3)}^{\mathbf{P}}) = [p(1)+p(2)] I_{\diamond}(\mathcal{M}_{(1,2)}^{\mathbf{q}}), \text{ where } \mathbf{q} = \left(\frac{p(1)}{p(1)+p(2)}, \frac{p(2)}{p(1)+p(2)} \right)$$

Implications for Steering

Steering assemblage

$$\vec{\sigma}_{(1,2,\dots,m)} = (\sigma_1, \sigma_2, \dots, \sigma_m) \text{ with } \sigma_x = \{\sigma_{a|x}\}_a$$

$$\sigma_{a|x} \succeq 0 \ \forall \ a, x, \quad \sum_a \sigma_{a|x} = \sum_a \sigma_{a|x'} = \rho_B \ \forall \ x, x'$$

Local hidden-state model (LHS)

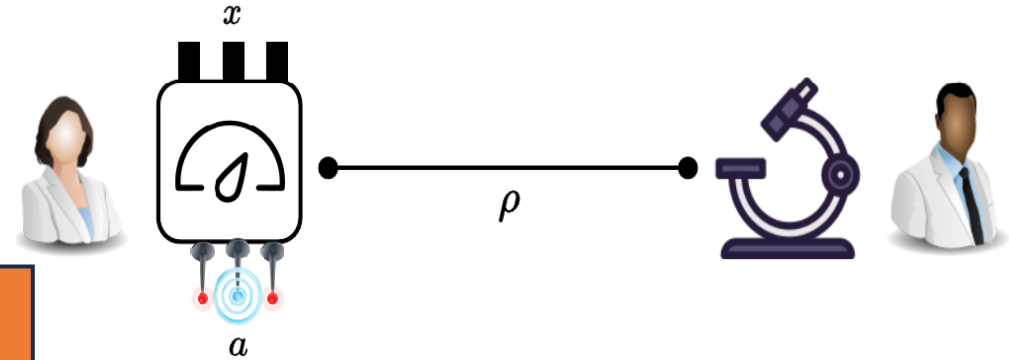
$$\sigma_{a|x} = \sum_{\lambda} p(a|x, \lambda) \sigma_{\lambda}$$

**Failing of a LHS
explanation certifies
entanglement and
measurement
incompatibility**

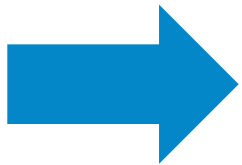
Quantification

$$S(\vec{\sigma}) = \min_{\vec{\tau} \in \text{CLHS}} \frac{1}{2} \sum_{a,x} \frac{1}{m} \|\sigma_{a|x} - \tau_{a|x}\|_1$$

Ku et al. PRA 97, 022338 (2018)



$$\sigma_{a|x} = \text{Tr}_A[(M_{a|x} \otimes \mathbb{1})\rho]$$



**Exactly the same formalism as in
the incompatibility case, up to a
no-signaling constraint!**

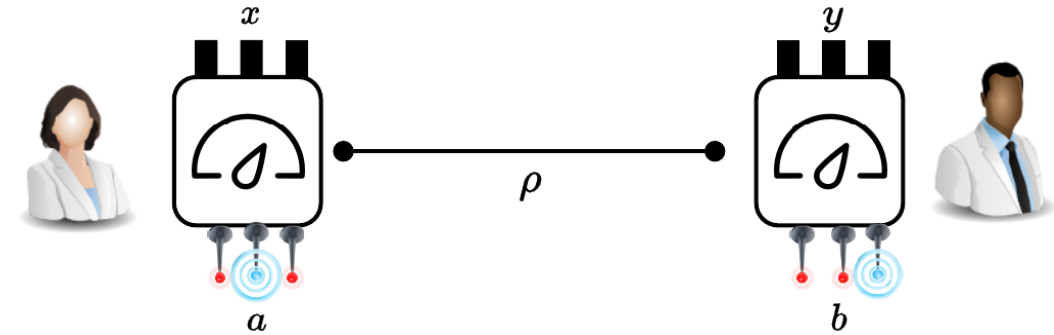
Implications for Nonlocality

Behaviour

$$\mathbf{q}_{(1,2,3)} = [q(ab|1y), \dots, q(ab|3y)]$$

Local hidden-variable model (LHV)

$$q(a, b|x, y) = \sum_{\lambda} \pi(\lambda) p_A(a|x, \lambda) p_B(b|y, \lambda)$$



$$q(ab|xy) = \text{Tr}[(A_{a|x} \otimes B_{b|y})\rho]$$

Quantification

$$N(\mathbf{q}) = \frac{1}{2} \min_{\mathbf{t} \in \text{CLHV}} \sum_{a,b,x,y} \frac{1}{m_A m_B} |q(a, b|x, y) - t(a, b|x, y)|$$

Bruto et al. PRA 97, 022111 (2018)

$$N(\mathbf{q}_{(1,2,3)}) \leq \frac{2}{3} N(\mathbf{q}_{(1,2)}) + N(\mathbf{q}_{(1,2,3)}^{\#(1,2)})$$

$$N(\mathbf{q}_{(1,2,3)}) \leq \frac{1}{3} [N(\mathbf{q}_{(1,2)}) + N(\mathbf{q}_{(1,3)}) + N(\mathbf{q}_{(2,3)})] + N(\mathbf{t}), \quad \mathbf{t} = \mathbf{q}_{(1,2)}^{\#} \sharp \mathbf{q}_{(1,3)}^{\#} \sharp \mathbf{q}_{(2,3)}^{\#}$$

Same
formalism

Does Nonlocality behave differently?

$$N(\mathbf{q}_{(1,2,3)}) \leq \underbrace{\frac{1}{3}[N(\mathbf{q}_{(1,2)}) + N(\mathbf{q}_{(1,3)}) + N(\mathbf{q}_{(2,3)})]}_{\text{Average subset nonlocality}} + N(\mathbf{t}), \quad \mathbf{t} = \mathbf{q}_{(1,2)}^\# \oplus \mathbf{q}_{(1,3)}^\# \oplus \mathbf{q}_{(2,3)}^\#$$

Consider dichotomic Bell scenario with 3 Settings for Alice and 2 for Bob

$$\text{CHSH}_{(i,j)} := \langle A_i \otimes B_1 \rangle + \langle A_i \otimes B_2 \rangle + \langle A_j \otimes B_1 \rangle - \langle A_j \otimes B_2 \rangle \leq 2, \quad (i,j) \in \{(1,2), (1,3), (2,3)\}$$

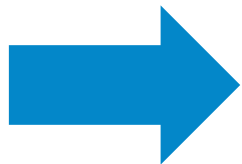
$$\text{CHSH}_{(1,2,3)} := \frac{1}{3}[\text{CHSH}_{(1,2)} + \text{CHSH}_{(1,3)} + \text{CHSH}_{(2,3)}] \stackrel{\text{LHV}}{\leq} 2 \stackrel{\text{Q}}{\leq} \frac{4\sqrt{2}+2}{3} < 2\sqrt{2}$$

Open Question:

Does there exist a better nonlocal game than CHSH for two outcome correlations?

Brito et al. PRA 97, 022111 (2018)

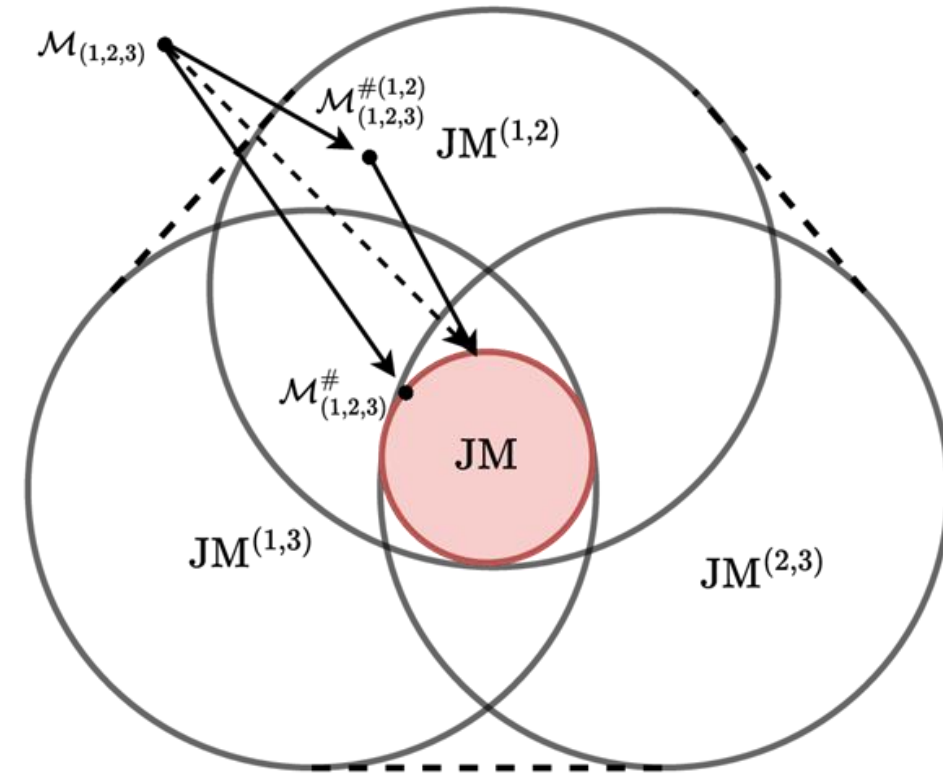
Araújo et al. Quantum 4, 353 (2020)



Average subset nonlocality is strictly smaller than maximal nonlocality of the subsets!

Summary

- Incompatibility of multiple measurements has a rich structure
- Distance-based approach is well suited to study structures
- Tight bounds on the incompatibility depending on properties of subsets
- Tight bounds on the incompatibility gain



4-8 November

<https://yqis24.sciencesconf.org/>

<https://easychair.org/cfp/YQIS2024>

Thank you!

$\langle \Psi | \text{QIS} \rangle 2024$

